

Reviewer Pack — Minimal Order of Twistor Space and Communication Complexity ...

Entry 85735569deb7 · INCONCLUSIVE

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Minimal Order of Twistor Space and Communication Complexity Rank Correlation

Entry ID: 85735569deb7 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-02 02:32:51 UTC

1. Conjecture

Field A (mathematical branch): Twistor Theory **Field B** (complexity object): Communication Complexity

Statement:

The minimal order of the twistor space associated with a given Boolean function φ is linearly correlated with its communication complexity rank $c(\varphi)$, such that $\text{min_order}(\text{TwistorSpace}(\varphi)) = \Theta(c(\varphi))$.

Rationale (proposer's reasoning):

Twistor theory, originally developed in theoretical physics, provides a geometric framework for understanding complex phenomena. Its application to communication complexity could reveal a novel approach to understanding the intrinsic complexity of communication tasks. The minimal order of the twistor space might serve as an invariant that captures the structure of Boolean functions in a way that is relevant to their communication complexity.

Taxonomy category: Twistor Theory × Communication Complexity (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 6f42170cf90b545b

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the Pearson correlation coefficient between $\text{min_order}(\text{TwistorSpace}(\varphi))$ and $c(\varphi)$ is ≥ 0.7 for all generated Boolean functions φ , with a p-value ≤ 0.05 across 30 independent experiments.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 2 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - 'Twistor Theory' AND 'Communication Complexity' AND 'minimal order' - 'twistor space' AND 'communication complexity rank' AND correlation - correlation between 'minimal order of twistor space' and 'communication complexity'

Top relevant hits considered: - [s2:202773104af18c35f559dae75a8d3c8b78be3660] Physician Order Entry Systems Should Be Integrated - [s2:8434e21e3437058af1808065c41e71fcbc963577] Dynamics of N-fold Integration of Disparate Cognitive Modalities

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=1.7s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def communication_complexity_rank(phi):
        n = len(phi)
        rank = 0
        for i in range(n):
            if all(phi[j] == phi[j + 2**i] for j in range(2**(n - i) - 1)):
                rank += 1
        return rank

    def twistor_space_order(phi):
        n = len(phi)
        order = 0
        for i in range(n):
            if all(phi[j] == phi[j + 2**i] for j in range(2**(n - i) - 1)):
                order += 1
        return order

    def pearson_correlation(x, y):
        n = len(x)
        mean_x = sum(x) / n
        mean_y = sum(y) / n
        cov = sum((x[i] - mean_x) * (y[i] - mean_y) for i in range(n)) / n
```

```

std_x = math.sqrt(sum((x[i] - mean_x)**2 for i in range(n)) / n)
std_y = math.sqrt(sum((y[i] - mean_y)**2 for i in range(n)) / n)
return cov / (std_x * std_y)

results = []
for _ in range(30):
    n = random.choice([5, 10, 15, 20, 30, 40])
    phi = generate_boolean_function(n)
    c_phi = communication_complexity_rank(phi)
    order_twistor_space = twistor_space_order(phi)
    results.append((order_twistor_space, c_phi))

if not results:
    return {
        "metric_name": "Pearson Correlation",
        "metric_value": None,
        "instances_tested": 0,
        "n_max": 0,
        "conjecture_holds": False,
        "counterexample": "No instances generated"
    }

x, y = zip(*results)
correlation = pearson_correlation(x, y)
return {
    "metric_name": "Pearson Correlation",
    "metric_value": correlation,
    "instances_tested": len(results),
    "n_max": max(len(phi) for phi, _ in results),
    "conjecture_holds": correlation >= 0.7,
    "counterexample": ""
}

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] if len(sys.argv) > 1 else [2**i + 3 for i in range(5, 8)]

    all_results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        all_results.append(result)

    mean_value = sum(r["metric_value"] for r in all_results if r["metric_value"] is not None) / len(all_results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in all_results if r["metric_value"] is not None) / len(all_results))
    support_fraction = sum(1 for r in all_results if r["conjecture_holds"]) / len(all_results)

    if support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in all_results):
        counterexample = next(r["counterexample"] for r in all_results if not r["conjecture_holds"])
        first_failing_seed = next(r["seed"] for r in all_results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE insufficient_data")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_69f2995f.py", line 83, in <module>
    result = run_trial(seed)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_69f2995f.py", line 53, in run_trial
    c_phi = communication_complexity_rank(phi)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_69f2995f.py", line 28, in communication
    if all(phi[j] == phi[j + 2*i] for j in range(2*(n - i) - 1)):
       ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_69f2995f.py", line 28, in <genexpr>
    if all(phi[j] == phi[j + 2*i] for j in range(2*(n - i) - 1)):
       ~~~~~
```

IndexError: list index out of range

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which means it did not complete the required experiments to evaluate the conjecture. | next: Re-run the test with proper error handling and ensure that it completes all necessary computations to verify the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13417
2	preregistration	ollama_remote	glm4:latest	0	0	23601
3	novelty	ollama_remote	glm4:latest	0	0	20292
4	novelty	ollama_remote	glm4:latest	0	0	14678
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16421
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10024
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12770
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	25986
9	judge	ollama_remote	glm4:latest	0	0	11072

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 148262 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/85735569deb7.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/85735569deb7.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/85735569deb7.tar.gz (if generated)